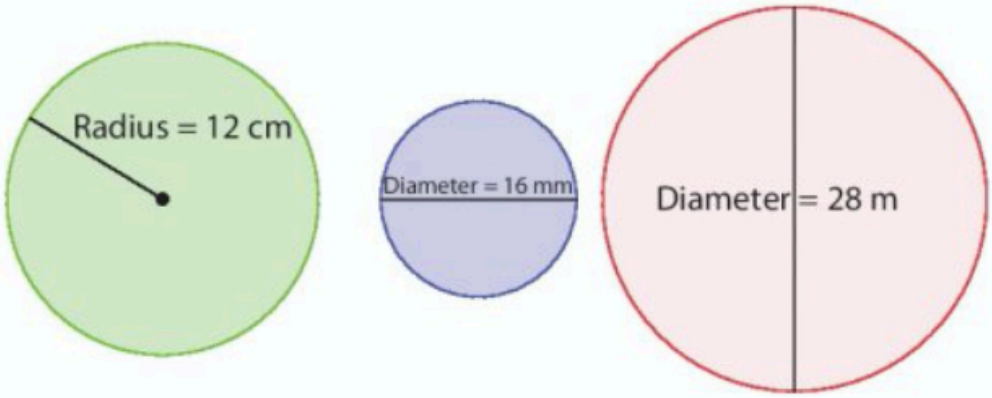
Exercise 5D

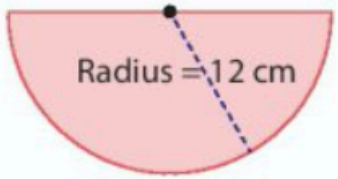
1. Use the π key on the calculator to find the area of the three discs below. Give each answer correct to 1 place of decimals:



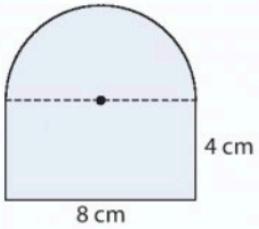
2. Find: (i) the area

(ii) the perimeter of this semi-circular disc.

Take $\pi = \frac{22}{7}$ in your calculations, giving your answers correct to 3 significant figures.

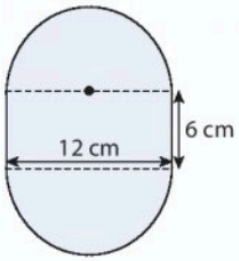


3. Find (i) the area
(ii) the perimeter of the given figure.
Give each answer correct to one decimal place.
Take $\pi = \frac{22}{7}$ in your calculations

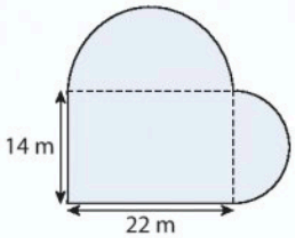


4. Each of the following figures has semicircular ends.
Find (i) the area (ii) the perimeter of each figure, taking $\pi = 3.14$ and giving your answer correct to 1 decimal place:

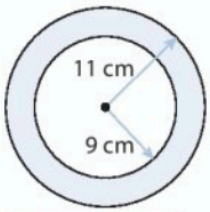
(i)



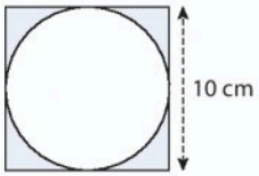
(ii)



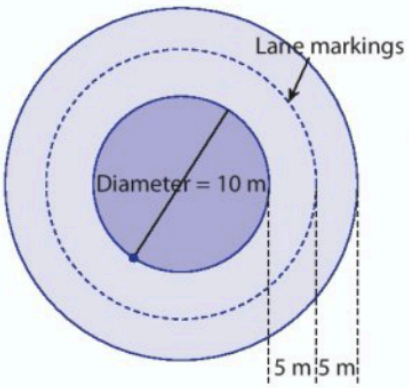
5. The diagram shows a cardboard ring. The inner radius is 9 cm.
The outer radius is 11 cm.
Calculate the area of the ring in cm^2, correct to the nearest whole number.
(Take $\pi = 3.14$ in your calculations)



6. A circle is drawn inside a square, as shown.
The square has side of length 10 cm.
Calculate the area of the shaded region, correct to one decimal places.
(Take $\pi = 3.14$ in your calculations)

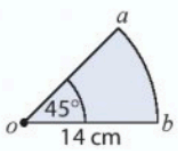


7. A roundabout (circular) of diameter 10 m has two lanes around it. Each lane is 5 m wide.
Using the π key on the calculator find how much tarmac is needed to renew the surface of these two lanes.
Give your answer correct to 2 significant figures.
Estimate the cost of this repair if:
(i) it costs €75 per m^2 to lay the tar and
(ii) €25 per metre to paint lane markings, answers correct to the nearest euro.

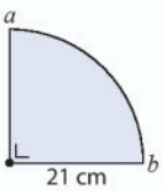
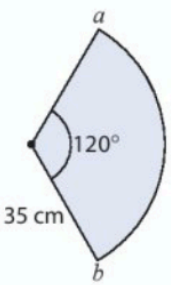


8. In the following diagrams find:
(i) the fraction of a disc represented by each figure
(ii) the area of each shaded sector
(iii) the length of the arc ***ab*** in each figure
(iv) the length of the perimeter of each figure.
Take $\pi = 3.14$ and give your answers correct to the nearest whole number.

(i)



(ii)



(iii)